

1. 求下列幂级数的收敛半径:

$$(1) \sum_{n=1}^{\infty} \frac{x^n}{n 2^n}; \quad (2) \sum_{n=1}^{\infty} \frac{n^k}{n!} x^n; \quad (3) \sum_{n=1}^{\infty} \frac{n!}{n^n} x^n; \quad (4) \sum_{n=1}^{\infty} \frac{(n!)^2}{(2n)!} x^n.$$

$$1. (1) \lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} = \lim_{n \rightarrow \infty} \frac{n \cdot 2^n}{(n+1) \cdot 2^{n+1}} = \frac{1}{2}, \quad R=2$$

$$(2) \lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} = \lim_{n \rightarrow \infty} \left(\frac{n+1}{n}\right)^k \cdot \frac{1}{n+1} = 0 \quad \therefore R=+\infty$$

$$(3) \lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} = \lim_{n \rightarrow \infty} (n+1) \frac{n^n}{(n+1)^{n+1}} = \lim_{n \rightarrow \infty} \left(\frac{n}{n+1}\right)^n = \frac{1}{e}, \quad R=e$$

$$(4) \lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} = \lim_{n \rightarrow \infty} \frac{(n+1)!(n+1)!}{(2n+2)!} \cdot \frac{(2n)!}{(n!)^2} = \lim_{n \rightarrow \infty} \frac{(n+1)^2}{2(2n+1)} = \lim_{n \rightarrow \infty} \frac{n+1}{2(2n+1)} = \frac{1}{4}, \quad R=4$$

2. 求下列幂级数的收敛区间与收敛域:

$$(1) x + \frac{x^2}{\sqrt[3]{2}} + \frac{x^3}{\sqrt[3]{3}} + \dots;$$

$$(2) 1 + \frac{x}{a} + \frac{x^2}{2a^2} + \dots + \frac{x^n}{na^n} + \dots \quad (a > 0);$$

$$(3) x - \frac{x^3}{3 \cdot 3!} + \frac{x^5}{5 \cdot 5!} - \dots + (-1)^n \frac{x^{2n+1}}{(2n+1) \cdot (2n+1)!} + \dots;$$

$$(4) \sum_{n=1}^{\infty} \frac{n}{n+1} \left(\frac{x}{2}\right)^n;$$

$$(5) x + x^3 + x^5 + \dots + x^{2n+1} + \dots;$$

$$(6) \sum_{n=1}^{\infty} (3^{-n} + 5^{-n}) x^n;$$

$$(7) \sum_{n=1}^{\infty} \left(\frac{1}{n} + e^{-n}\right) x^n;$$

$$(4) \left| \frac{a_{n+1}}{a_n} \right| = \frac{n+1}{(n+2) \cdot 2^{n+1}} \cdot \frac{2^n (n+1)}{n} = \frac{(n+1)^2}{2n(n+2)} \quad \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = \frac{1}{2}, \quad \therefore R=2$$

$x=2$ 时, 发散 $x=-2$ 时, ~~收敛~~ 发散

$\therefore$ 收敛域: $(-2, 2)$ 区间: $(-2, 2)$

$$(5) a_n = 3^{-n} + 5^{-n} = 5^{-n} \cdot \left(\left(\frac{3}{5}\right)^{-n} + 1\right) = \frac{1}{5^n} \left[\left(\frac{5}{3}\right)^n + 1\right] = \frac{1}{3^n} \left[\left(\frac{3}{5}\right)^n + 1\right]$$

$$\lim_{n \rightarrow \infty} n \sqrt[n]{a_n} = \frac{1}{3} \quad \therefore R=3$$

$x=3$ 时, 易知发散; $x=-3$ 时, $u_n = (-1)^n + (-1)^n \cdot \left(\frac{3}{5}\right)^n$, $\lim_{n \rightarrow \infty} |u_n| = 1$, 发散

$\therefore$ 收敛区间均为 $(-3, 3)$

$$(5) \left| \frac{u_{n+1}}{u_n} \right| = |x|^2 < 1, \quad x \in (-1, 1), \quad x=1 \text{时, 发散}; \quad x=-1 \text{时, 发散}$$

$\therefore$ 区间/域为 $(-1, 1)$

$$(7) \frac{1}{n} + \frac{1}{e^n} = \frac{e^n + 1}{n e^n}, \quad n \sqrt[n]{a_n} = \frac{1}{e} \cdot n \sqrt[n]{\frac{e^n + 1}{n}}, \quad \lim_{n \rightarrow \infty} n \sqrt[n]{a_n} = 1$$

$x=1$ 时, 发散; $x=-1$ 时, 单调且 $\rightarrow 0$, 收敛

$\therefore$ 收敛区间: $(-1, 1)$ 收敛域: $[-1, 1)$

$$2. (1) \sum_{n=1}^{\infty} \frac{x^n}{\sqrt[3]{n}}, \quad \frac{a_{n+1}}{a_n} = \sqrt[3]{\frac{n}{n+1}}, \quad \lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} = 1, \quad \therefore R=1,$$

$x=1$ 时, 易知发散; $x=-1$ 时, $|u_n| = \frac{1}{\sqrt[3]{n}}$ 单调且趋于0, 收敛

收敛区间: $(-1, 1)$ 收敛域: $[-1, 1)$

$$(2) \frac{a_{n+1}}{a_n} = \frac{n a^n}{(n+1) a^{n+1}} = \frac{n}{n+1} \cdot \frac{1}{a} \quad \lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} = \frac{1}{a}, \quad \therefore R=a$$

$x=a$ 时, $u_n = \frac{1}{n}$, 发散; $x=-a$ 时, $u_n = \frac{(-1)^n}{n}$, 收敛

区间: $(-a, a)$ 域: $[-a, a)$

$$(3) \lim_{n \rightarrow \infty} \left| \frac{u_{n+1}}{u_n} \right| = \lim_{n \rightarrow \infty} \frac{(2n+1) \cdot (2n+1)!}{(2n+3) \cdot (2n+3)!} x^2 = \lim_{n \rightarrow \infty} \frac{2n+1}{(2n+2)(2n+3)^2} x^2 = 0$$

$\therefore R=+\infty$, 收敛区间与域均为 $\mathbb{R}$

$$(8) \sum_{n=1}^{\infty} (3^n + 5^n) x^n; \quad (9) \sum_{n=1}^{\infty} \left(1 + \frac{1}{2} + \cdots + \frac{1}{n}\right) x^n;$$

$$(10) \sum_{n=1}^{\infty} \frac{(2n)!!}{(2n+1)!!} x^n \text{ (提示: 为讨论在 } x = \pm 1 \text{ 处的敛散性, 请见习题 10.3 第 1 题中(11) 小题);}$$

$$(11) \sum_{n=1}^{\infty} \frac{n! (x-2)^n}{n^n} \text{ (提示: 为证明在 } x = 2 \pm e \text{ 处级数发散, 请注意 } \frac{e}{(1+1/n)^n})}$$

$$(11). \text{ 令 } t = x-2, \quad \sum_{n=1}^{\infty} \frac{n!}{n^n} t^n, \quad \frac{u_{n+1}}{u_n} = (n+1) \cdot \frac{n^n}{(n+1)^{n+1}} = \left(\frac{n}{n+1}\right)^n$$

$$\lim_{n \rightarrow \infty} \left| \frac{u_{n+1}}{u_n} \right| = \frac{1}{e}, \quad \therefore R = e, \quad \therefore x \text{ 的收敛区间为 } (2-e, 2+e)$$

$$t = e \text{ 时, } \frac{u_{n+1}}{u_n} = e \cdot (n+1) \cdot \frac{n^n}{(n+1)^{n+1}} = e \cdot \left(\frac{n}{n+1}\right)^n = \frac{e}{(1+\frac{1}{n})^n} > 1, \text{ 单增}$$

发散

$$t = -e \text{ 时, } u_n = (-1)^n \cdot \frac{n! e^n}{n^n}, \text{ 同理亦发散}$$

$$(8) \lim_{n \rightarrow \infty} \sqrt[n]{a_n} = 5, \quad R = \frac{1}{5} \quad x = \frac{1}{5} \text{ 时, 发散; } x = -\frac{1}{5} \text{ 时, 发散}$$

$\therefore$ 区间与端点均为 $(-\frac{1}{5}, \frac{1}{5})$

$$(9) \lim_{n \rightarrow \infty} \frac{\sum_{k=1}^{n+1} \frac{1}{k}}{\sum_{k=1}^n \frac{1}{k}} = 1, \quad x = \pm 1 \text{ 时均发散} \quad \text{故收敛区间/端点为 } (-1, 1)$$

$$(10) \lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} = \lim_{n \rightarrow \infty} \frac{(2n+2)!!}{(2n+3)!!} \cdot \frac{(2n+1)!!}{(2n)!!} = \lim_{n \rightarrow \infty} \frac{2n+2}{2n+3} = 1$$

$$\textcircled{1} x=1 \text{ 时, } \therefore \frac{n-1}{n} < \frac{n}{n+1} \quad \frac{n}{n-1} > \frac{n+1}{n}$$

$$\therefore \frac{2 \times 4 \times \cdots \times 2n}{1 \times 3 \times 5 \times \cdots \times (2n+1)} > \frac{1}{2n+1} \cdot \frac{3 \times 5 \times \cdots \times (2n+1)}{2 \times 4 \times \cdots \times 2n} = \frac{1}{2n+1} \cdot \frac{(2n+1)!!}{(2n)!!}$$

$$\therefore u_n^2 > \frac{1}{2n+1}, \quad u_n > \frac{1}{\sqrt{2n+1}}, \quad \therefore \text{发散.}$$

$$\textcircled{2} x=-1 \text{ 时, } \therefore \frac{(2n)!!}{(2n+1)!!} < \frac{(2n+1)!!}{(2n)!!} \cdot \frac{1}{2n}, \quad \therefore |u_n| < \frac{1}{\sqrt{2n}}, \quad \therefore u_n \rightarrow 0.$$

又有 $|u_n| \downarrow$

故收敛

$\therefore$ 收敛域为 $[-1, 1)$, 区间为 $(-1, 1)$

3. 求下列幂级数的和函数:

$$(1) \sum_{n=0}^{\infty} (n+1)x^n; \quad (2) \sum_{n=1}^{\infty} (-1)^{n-1} (2n-1)x^{2n-2};$$

$$(3) \sum_{n=1}^{\infty} (-1)^{n-1} \frac{x^{n+1}}{n(n+1)}; \quad (4) \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n(2n-1)} x^{2n};$$

$$(5) \sum_{n=0}^{\infty} \frac{(2n+1)}{n!} x^{2n}; \quad (6) \sum_{n=1}^{\infty} \frac{2n-1}{2^n} x^{2n-2};$$

$$(7) \sum_{n=1}^{\infty} \frac{x^n}{n(n+1)}.$$

$$3. (1) (n+1)x^n = (x^{n+1})'$$

$$\sum_{n=0}^{\infty} (n+1)x^n = \sum_{n=0}^{\infty} (x^{n+1})' = \left(\sum_{n=0}^{\infty} x^{n+1} \right)' = \left(\frac{x}{1-x} \right)' = \frac{1-x+x}{(1-x)^2} = \frac{1}{(1-x)^2}, x \in (-1, 1)$$

$$(2) (-1)^{n-1} (x^{2n-1})' = [(-1)^{n-1} x^{2n-1}]', \text{ 又有 } R=1, -1 < x < 1$$

$$\text{原式} = \left(\sum_{n=1}^{\infty} (-1)^{n-1} x^{2n-1} \right)', \quad \sum_{n=1}^{\infty} (-1)^{n-1} x^{2n-1} = x - x^3 + \dots = \lim_{n \rightarrow \infty} \frac{x(1 - (-x^2)^n)}{1 + x^2} = \frac{x}{1+x^2}$$

$$\therefore \text{原式} = \left(\frac{x}{1+x^2} \right)' = \frac{1+x^2-2x^2}{(1+x^2)^2} = \frac{1-x^2}{(1+x^2)^2}$$

$$(3) S(x) = \sum_{n=1}^{\infty} (-1)^{n-1} \frac{x^{n+1}}{n(n+1)}, \quad R=1, -1 < x < 1$$

$$S'(x) = \sum_{n=1}^{\infty} (-1)^{n-1} \frac{x^n}{n}, \quad S''(x) = \sum_{n=1}^{\infty} (-1)^{n-1} x^{n-1} = \sum_{n=0}^{\infty} (-1)^n x^n = 1 - x + x^2 - \dots = \frac{1}{1+x}$$

$$S'(0) = 0, \quad S'(x) = \ln(x+1),$$

$$S(0) = 0, \quad S(x) = (x+1) \ln(x+1) - x$$

$$(4) S(x) = \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n(2n-1)} x^{2n}, \quad R=1, -1 < x < 1$$

$$S'(x) = \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{2n-1} \cdot 2x^{2n-1}, \quad S''(x) = \sum_{n=1}^{\infty} (-1)^{n-1} \cdot 2x^{2n-2} = 2 \sum_{n=1}^{\infty} (-1)^{n-1} x^{2n-2} = \frac{2}{1+x^2}$$

$$S'(0) = 0,$$

$$S'(x) = 2 \arctan x$$

$$S(0) = 0,$$

$$S(x) = 2 \int_0^x \arctan t \, dt = 2 \cdot \left(x \arctan x - \int_0^x \frac{x}{1+x^2} dx \right) = 2x \arctan x - \ln(x^2+1)$$

$$(5) \sum_{n=0}^{\infty} \frac{(2n+1)}{n!} x^{2n}$$

$$\frac{a_{n+1}}{a_n} = \frac{2n+3}{(n+1)!} \cdot \frac{n!}{2n+1} = \frac{2n+3}{(n+1)(2n+1)} \rightarrow 0, \quad R = +\infty$$

$$\text{原式} = \left(\sum_{n=0}^{\infty} \frac{x^{2n+1}}{n!} \right)', \quad \sum_{n=0}^{\infty} \frac{x^{2n+1}}{n!} = x + x^3 + \frac{x^5}{2!} + \dots + \frac{x^{2n+1}}{n!} + \dots = x \cdot e^{x^2}$$

$$\therefore \text{原式} = \frac{1}{2} e^{x^2}$$

$$(6) R = \sqrt{2}, \quad x \in (-\sqrt{2}, \sqrt{2}) \quad \left(\sum_{n=1}^{\infty} \frac{x^{2n-1}}{2^n} \right)', \quad \sum_{n=1}^{\infty} \frac{x^{2n-1}}{2^n} = \lim_{n \rightarrow \infty} \frac{\frac{x}{2} (1 - (\frac{x^2}{2})^n)}{1 - \frac{x^2}{2}} = \frac{x}{2-x^2}$$

$$S(0) = 0, \quad \therefore S(x) = \int_0^x \frac{t}{2-t^2} dt = -\frac{1}{2} \ln|2-x^2|$$

$$(7) \quad \lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} = 1, \quad x=1, \quad \text{原式} = \sum_{n=1}^{\infty} \frac{x^n}{n} - \sum_{n=1}^{\infty} \frac{x^n}{n+1} \\ = \ln(1-x) - \sum_{n=1}^{\infty} \frac{x^n}{n+1}$$

$$\sum_{n=1}^{\infty} \frac{x^n}{n+1} = \frac{1}{x} \sum_{n=1}^{\infty} \frac{x^{n+1}}{n+1}, \quad \sum_{n=1}^{\infty} \frac{x^{n+1}}{n+1} \text{ 记为 } S(x). \\ S'(x) = \sum_{n=1}^{\infty} x^n = x + x^2 + \dots + x^n + \dots = \lim_{n \rightarrow \infty} \frac{x(1-x^{n+1})}{1-x} = \frac{x}{1-x} \\ S(x) = \int \frac{x}{1-x} dx = \int (-1 + \frac{1}{1-x}) dx = -x + \ln(1-x)$$

$$\therefore \sum_{n=1}^{\infty} \frac{x^n}{n+1} = -1 + \frac{\ln(1-x)}{x}$$

$$\therefore \text{原式} = (1 + \frac{1}{x}) \ln(1-x) + 1 \quad (x \neq 1 \text{ 且 } x \neq 0)$$

$$x=0 \text{ 时, 原式} = 0$$

$$x=1 \text{ 时, 原式} = \sum_{n=1}^{\infty} (\frac{1}{n} - \frac{1}{n+1}) = 1$$

$$\therefore \text{原式} = \begin{cases} (1 + \frac{1}{x}) \ln(1-x) + 1, & x \neq 1, x \neq 0 \\ 0, & x = 0 \\ 1, & x = 1 \end{cases} \quad \text{, } 1 - 2\ln 2, x = -1$$

$x=-1$ 要单独讨论, 因为数列只在 $(-1, 1)$ 上一致收敛

$$x=-1, \quad \sum_{n=1}^{\infty} \frac{(-1)^n}{n(n+1)} = \sum_{n=1}^{\infty} (-1)^n \cdot (\frac{1}{n} - \frac{1}{n+1}) = -1 + \frac{1}{2} + \frac{1}{2} - \frac{1}{3} - \frac{1}{3} + \dots \\ = -1 + 2 \cdot \sum_{n=2}^{\infty} \frac{(-1)^n}{n}$$

$$S(x) = \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n+1} x^{n+1}, \quad S'(x) = \sum_{n=1}^{\infty} (-1)^{n+1} \cdot x^n = \lim_{n \rightarrow \infty} (x - x^2 + \dots + (-1)^{n+1} x^n) \\ = \lim_{n \rightarrow \infty} \frac{x \cdot (1 - (-x)^{n+1})}{1+x} = \frac{x}{1+x} = 1 - \frac{1}{1+x}$$

$$S(x) = x - \ln(1+x), \quad S(1) = 1 - \ln 2, \quad \text{原式} = 1 - 2\ln 2.$$